

Analysis and Visualisation

Week 5 · From numbers to evidence: summary statistics, pivot tables and honest charts

INF131 Technology and Artificial Intelligence · Faculty of Economics

Today's plan

1 Summary numbers

Totals, averages, percentages and growth rates

2 Tables and pivot tables

Summarise 960 rows in a few clicks

3 Choosing the right chart

Match the chart to the question, then remove clutter

4 Spotting misleading charts

Axis tricks, cherry-picking, 3-D and more

Lab: Excel 3 – tables, pivot tables, pivot charts, slicers, and fixing a badly designed chart

Analysis turns data into evidence for a decision



Question	Data	Summary	Chart	Decision
Where should we open the next store?	960 rows of monthly sales	Revenue growth by city	Bar chart, fastest city highlighted	Look closer at X

Takeaway: Always start from the question. The question decides about the data, fasibility of the study, context, and which number and which chart you need.

1 Summary numbers

Totals, averages, percentages, growth rates

Our dataset: X Market, 2024–2025

960 rows of data

5 cities

4 product categories

24 months

Month	City	Category	Channel	Revenue
Jan 2024	Tirana	Groceries	Store	€32,819
Jan 2024	Tirana	Groceries	Online	€3,174
Jan 2024	Tirana	Household	Store	€13,518
…	…	…	…	…
Dec 2025	Korçë	Clothing	Online	€965

Plus Units and Unit Price columns. One row = one city × category × channel × month. This dataset is all simulated in Python :)

Total revenue, both years

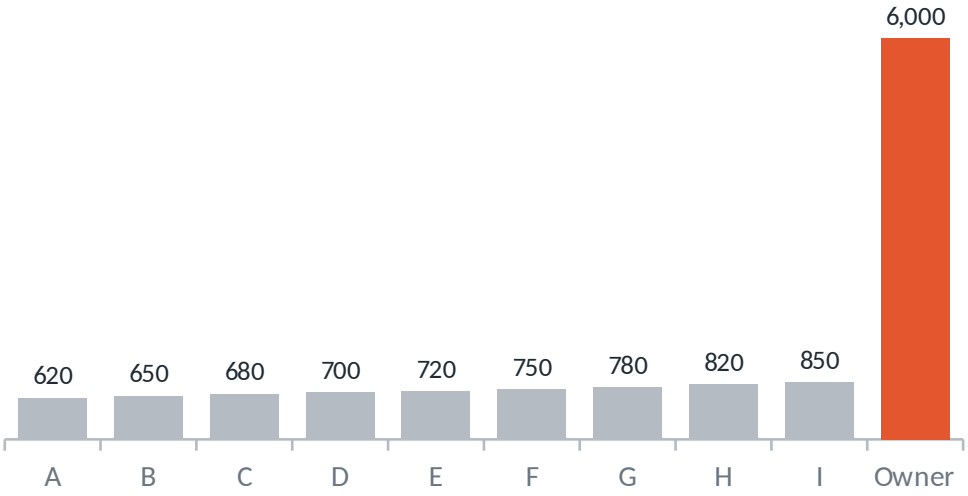
€5.18 million

`=SUM(Revenue)` `rows: =COUNT(Units)`

Takeaway 960 rows are too many to read. Summary numbers and charts make them readable.

Mean or median: which one is “typical”?

Salaries in a small firm, 10 employees (€ per month)



Mean (AVERAGE)

€1,257 pulled up by one big value

Median (MEDIAN)

€735 the middle person: more typical

Our data: mean row €5,396, median €3,010

Takeaway When a few values are very large (incomes, prices, sales), report the median too.

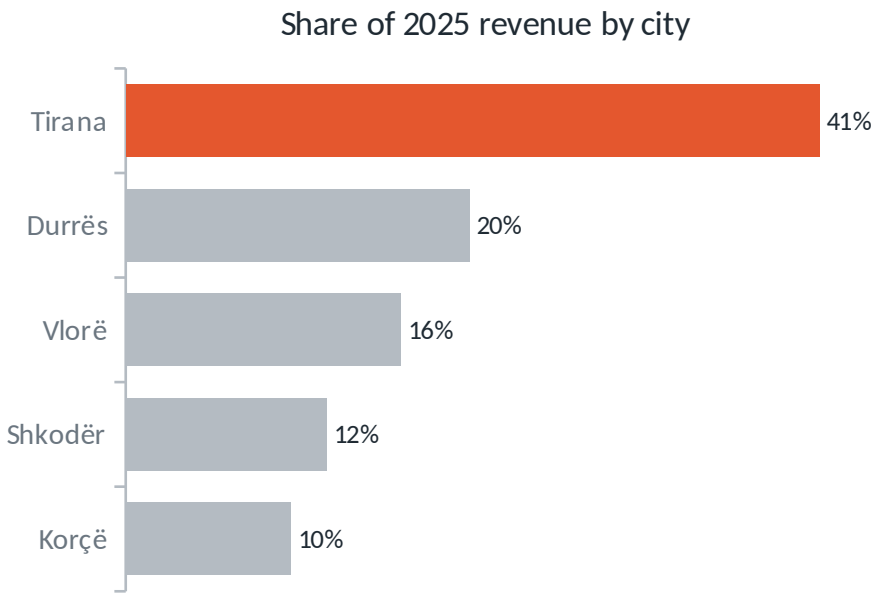
Percentage share: part ÷ whole

```
Share = part / whole
```

In Excel, lock the total with $ so you can copy the formula down:

```
=D25/$D$30
```

Then format the cell as %. 0.413 becomes 41.3%. All shares together = 100%.



Takeaway Tirana alone brings in about 41% of the chain's revenue.

Growth rate: how fast did it change?

```
Growth % = (new − old) / old
```

X Market, total revenue

2024	€2,457,780
2025	€2,722,221
Change	+€264,441
Growth	**+10.8%**

Watch out: ups and downs are not symmetric

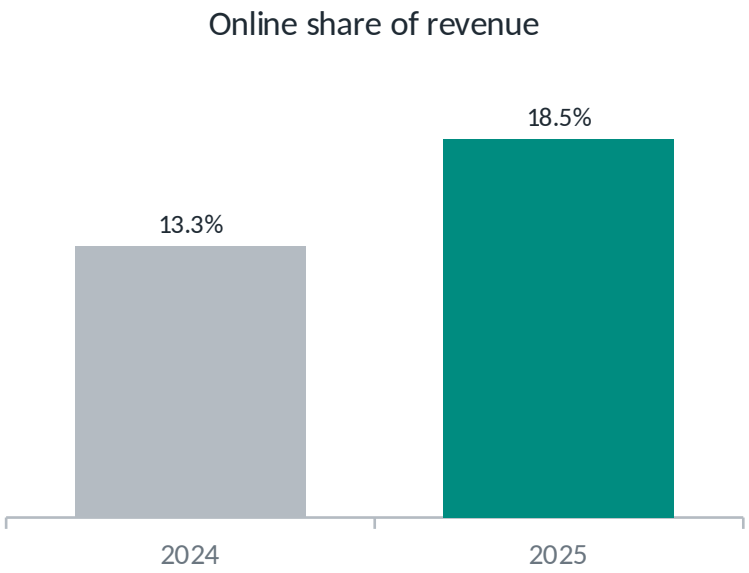
Price goes 80 → 100
(100 − 80) / 80 = +25%

Price goes back 100 → 80
(80 − 100) / 100 = −20%

Same €20, different %, because the starting point changed.

Takeaway Always divide by the starting (old) value, then format as %.

Percent vs percentage points



+5.2 percentage points

new share − old share · 18.5% − 13.3%

+39% (percent change)

(new − old) / old · (18.5 − 13.3) / 13.3

Same idea: the interest rate rises from 2% to 3% = +1 percentage point = +50%.

Takeaway When the number is already a %, say “percentage points” for the difference.

Quick check: try these in pairs

Revenue went from €100k to €120k. Growth rate?

Answer: +20%

A city sold €30k out of €120k in total. Its share?

Answer: 25%

Unemployment went from 6% to 9%. Describe the change two ways.

Answer: +3 pp, or +50%

Daily sales: 2, 5, 3, 26, 4. Mean or median: which is typical?

Answer: Median = 4 (mean = 8)

2

Tables and pivot tables

Summarise hundreds of rows in a few clicks

Step 1: turn your data into an Excel Table (Ctrl + T)

Grows automatically

New rows are included in formulas, charts and pivots

Calculated columns

Type =[@Units]*[@Price] once, it fills every row

Filter and sort buttons

Built into every header

A name

tblSales is easier to read than A1:I961

City ▾	Units ▾	Price ▾	Revenue ▾
Tirana	7,342	€4.47	€32,819
Tirana	710	€4.47	€3,174
Durrës	3,167	€4.54	€14,378
Vlorë	2,136	€4.54	€9,697
Korçë	1,983	€4.47	€8,864

```
Revenue: =[@Units]*[@[Unit Price]]
```

Takeaway Tidy data in a Table is the starting point for pivots, charts and all analysis that follows

Step 2: a pivot table answers “total of X by Y”

Rows

City

what goes down the side

Columns

Year

what goes across the top

Values

Sum of Revenue

the number to summarise

Filters / Slicers

Channel, Category

which rows to include

Sum of Revenue	2024	2025
Tirana	€1013k	€1125k
Durrës	€498k	€557k
Vlorë	€370k	€446k
…	…	…

Insert → PivotTable → drag fields into the four boxes

Takeaway No formulas needed: drag a field, and Excel groups and adds for you.

The finished pivot: revenue by city and year

Row Labels	2024	2025	Grand Total	Growth
Tirana	€1,012,563	€1,124,892	€2,137,456	+11.1%
Durrës	€498,186	€557,323	€1,055,509	+11.9%
Vlorë	€370,381	€446,364	€816,745	**+20.5%**
Shkodër	€311,028	€325,673	€636,701	+4.7%
Korçë	€265,622	€267,967	€533,590	+0.9%
Grand Total	**€2,457,780**	**€2,722,221**	**€5,180,000**	**+10.8%**

Growth column added for discussion; in Excel use Show Values As → % Difference From.

Try these in the pivot

- Sort largest to smallest
- Show Values As → % of Column Total
- Swap City and Year
- Put Category in Rows under City

Takeaway One pivot answers many questions: which city is biggest, which grew fastest.

Step 3: pivot charts and slicers make it interactive

Channel

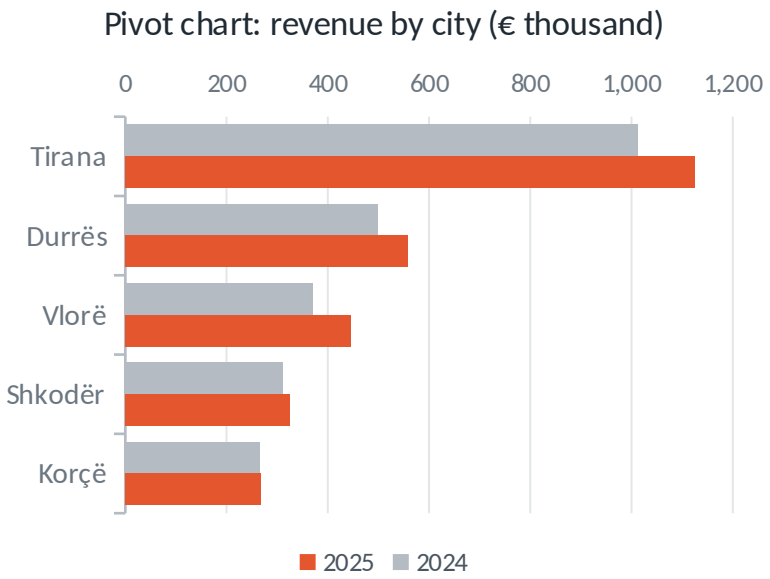
Online

Store

Category

Electronics

Groceries



How to

1. Click in the pivot
2. PivotTable Analyze → PivotChart
3. PivotTable Analyze → Insert Slicer
4. Click a slicer button: table and chart both update

Takeaway Slicers are filters you can click. Power BI works the same way.

3

Choosing the right chart

Start from the message, pick the form, remove clutter

What do you want to show? The job picks the chart

Compare categories

Bar / column chart

e.g. Revenue by city

Change over time

Line chart

e.g. Monthly revenue

Part of a whole

Pie (2–3 parts) or bar

e.g. Store vs online

Relationship

Scatter chart

e.g. Ad spend vs revenue

One key number

Big number, no chart

e.g. Growth +10.8%

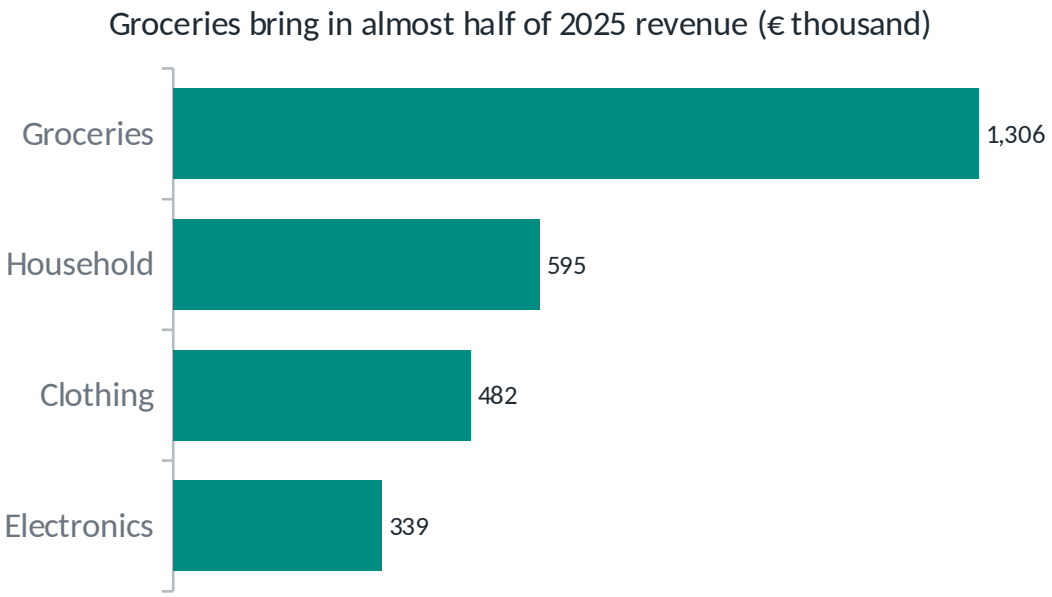
Exact values

Table

e.g. Price list, pivot

Finish the sentence first: “I want to show …”

Compare categories with a bar chart

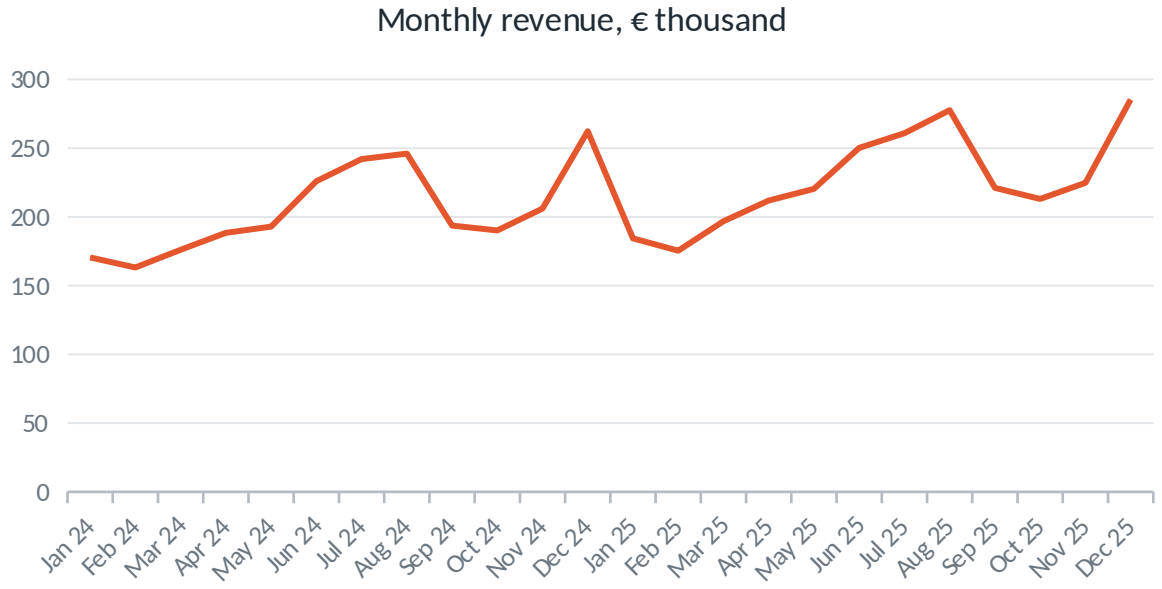


Good bar charts

- Axis starts at zero
- Sorted, biggest first
- Horizontal when labels are long
- One colour; values on the bars
- Title says the message

Takeaway Bars compare lengths, so the axis must always start at zero.

Show change over time with a line chart



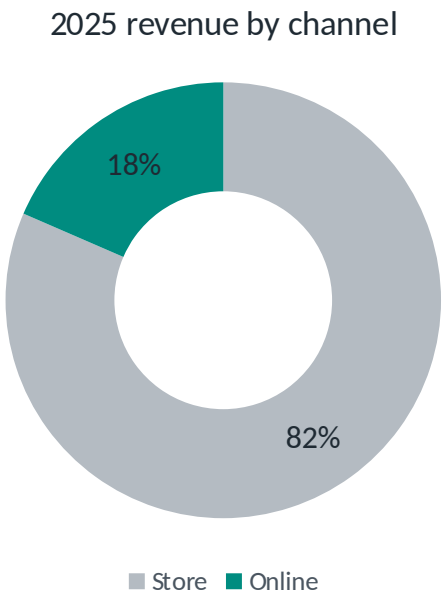
Time on the horizontal axis
Left = old, right = new

See the pattern
Peaks every summer and December

See the trend
2025 above 2024 in every month

Takeaway Time series? Use a line. Many points in time are clearer as a line than as bars.

Pie charts: only for a few parts of one whole



Pie is OK when…

- 2–3 slices only
- Parts add to one whole (100%)
- The message is “one big part”

Use a bar instead when…

- 5 or more slices
- Slices have similar sizes
- Data is over time
- Someone asks for 3-D

Takeaway If you need to read the labels to compare slices, a bar chart is much better.

Show a relationship with a scatter chart



Each dot = one store

More ad spend goes with more revenue: a positive relationship.

Correlation ≠ causation

Bigger stores may simply do both. A chart shows a link, not the reason.

Takeaway Two numbers per item? Scatter. Then ask what else could explain the pattern.

Sometimes the best chart is no chart

+10.8%

X Market revenue grew from 2024 to 2025, led by Vlorë (+21%).

One number → write it big

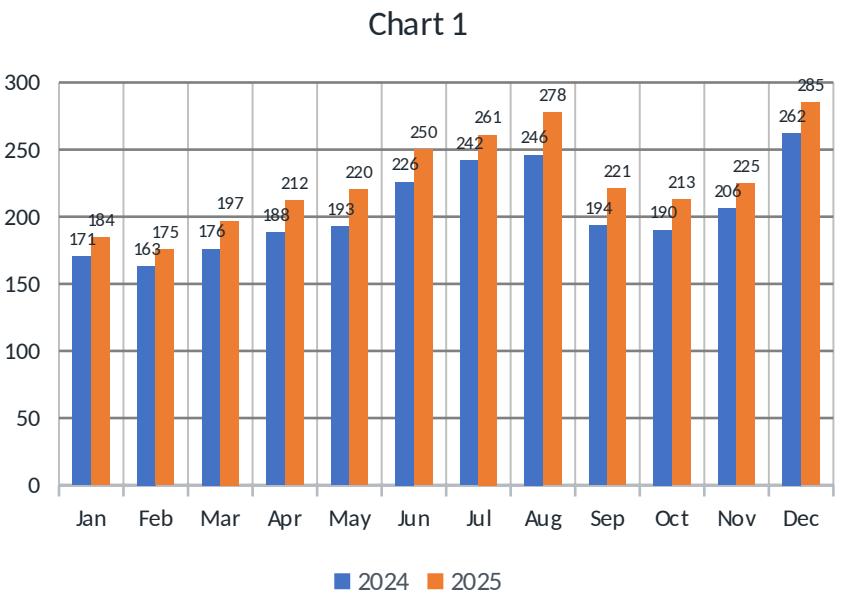
Category	Avg price 2025
Groceries	€4.63
Household	€12.40
Clothing	€28.85
Electronics	€98.05

Exact values to look up → use a table

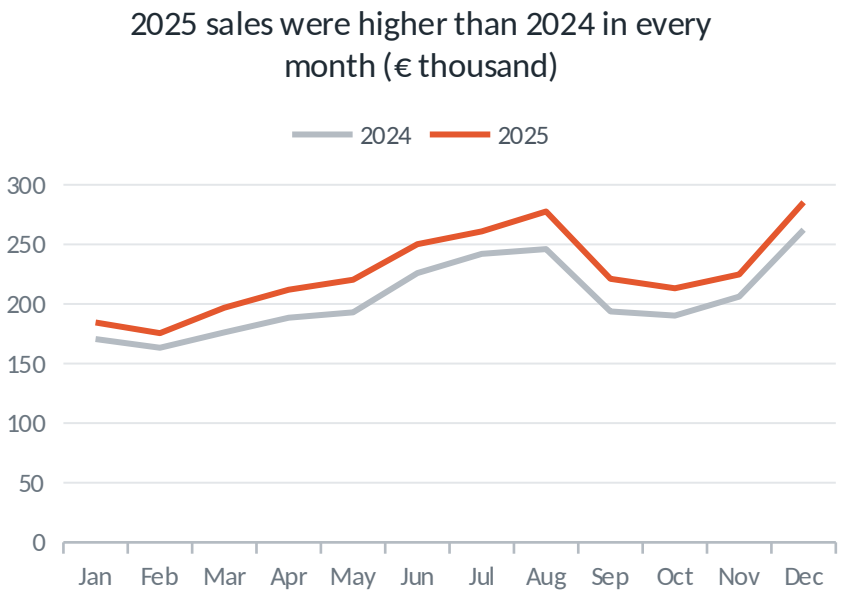
Takeaway Charts show shapes and comparisons. Big numbers and tables show exact values.

Remove clutter: every element must earn its place

Before

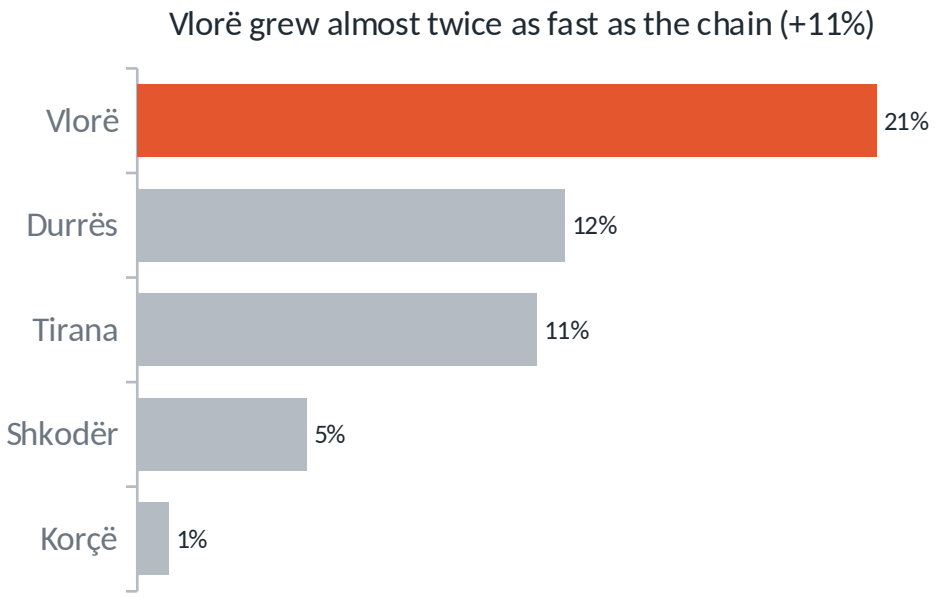


After



Takeaway Delete gridlines, extra labels and default colours. Keep what helps the message.

Focus attention: grey everything, colour one thing



Grey is the default

Most data is context

One accent colour

Only for what matters

Title = the message

Tell the reader what to see

Takeaway If everything is highlighted, nothing is. Pick one thing to colour.

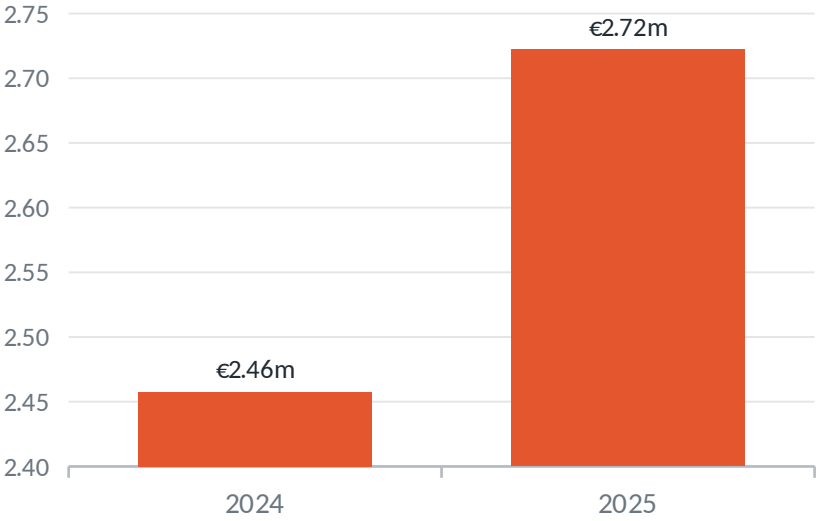
4

Spotting misleading charts

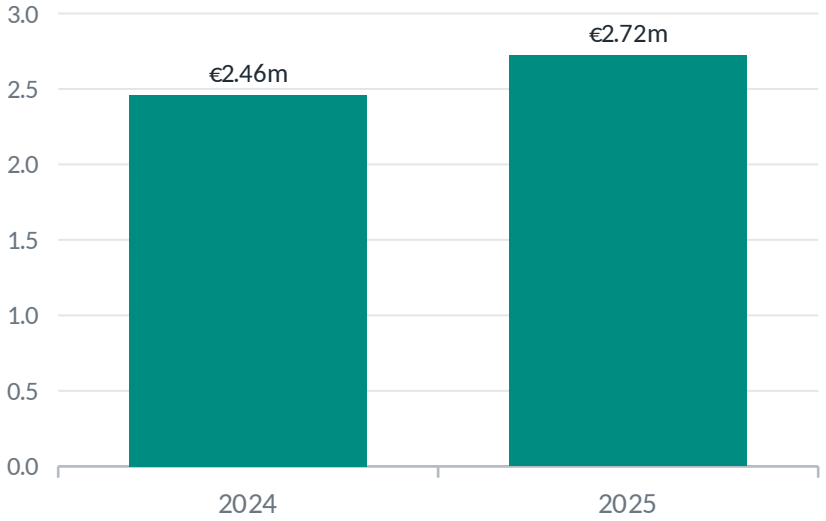
Correct numbers, wrong impression, media and Politicians love this!

Trick 1: the axis that does not start at zero

"Sales explode!" (axis starts at €2.4m)



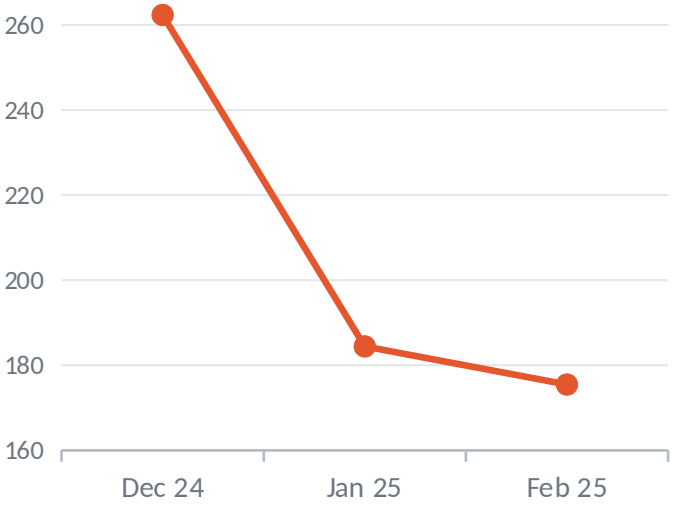
Honest: axis starts at zero (+11%)



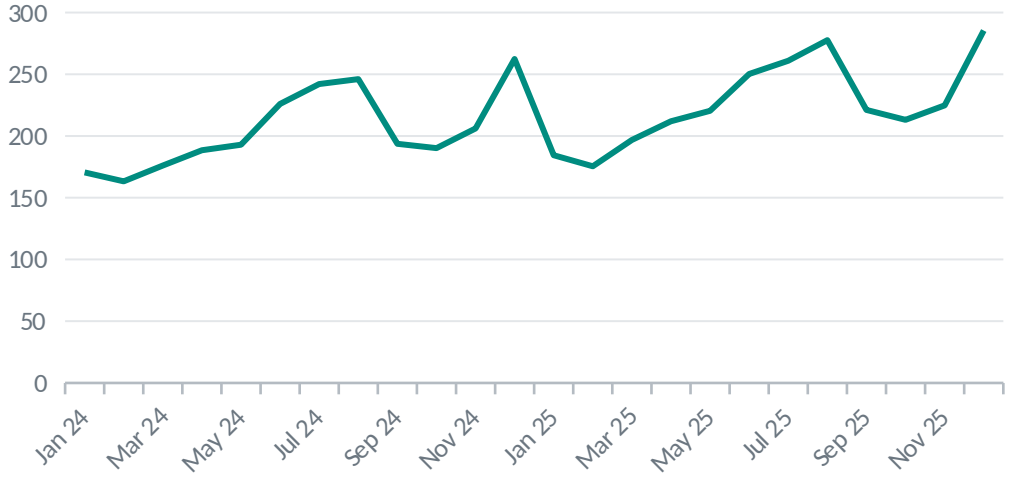
Takeaway Always read the axis first. Bars must start at zero.

Trick 2: cherry-picking the time window

"Sales collapse by 33%!"



Full picture: a normal January dip in a growing business (€ thousand)



Takeaway Ask why the chart starts and ends where it does. Look at a longer period.

More tricks to watch for

3-D effects

Front parts look bigger than they are

Two different y-axes

Can make any two lines look linked

Pie that adds to >100%

Parts that are not one whole

Pictures scaled by size

2× taller icon = 4× the area

No labels, units or source

You cannot check it

Wrong chart type

e.g. a pie chart for months

Takeaway A chart can use correct numbers and still give the wrong impression.

Checklist: before you share any chart

- What is my message? The title says it
- Right chart for the job (bar, line, scatter…)
- Bars start at zero; axes are labelled with units
- Sorted in a sensible order
- No 3-D, no extra gridlines, no default rainbow
- One accent colour for what matters
- Source and time period are stated
- It would still be honest without the title

Takeaway Use this list in today's lab, in your project, and on charts AI tools make for you.